

Introduction to the Markov chains

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Defining *state space* and *state values*

- Let S be a finite set with n elements, $S = \{x_1, x_2, \dots, x_n\}$.
- The set S is called the *state space* and $x_1, x_2, \dots, x_n$ are the state values.

Examples:

- State space for an economy:
 $S = \{\text{Normal Growth, Mild Recession, Severe recession}\}$
- State space for an agent: $S = \{\text{Employed, Unemployed}\}$

Markov chains: intro

A Markov chain $\{X_t\}$ on S is a sequence of random variables that have the **Markov property**:

For any date t and any state $y \in S$

$$P(X_{t+1} = y \mid X_t, X_{t-1}, \dots, X_0) = P(X_{t+1} = y \mid X_t)$$

This implies that:

- Past & future are conditionally independent given the present
- Knowing the current state is enough to know probabilities for future states.

Define a transition matrix P

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P_{ij} is called a *transition probability*.

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$$\mathbf{P} = \begin{bmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{bmatrix}$$

Example 2: MC through a directed graph lense

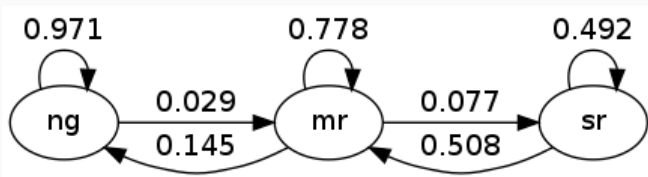
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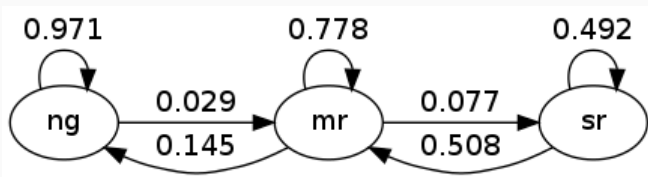
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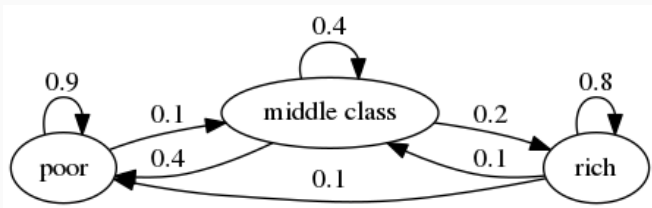
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$$P = \begin{bmatrix} 0.971 & 0.029 & 0 \\ 0.145 & 0.778 & 0.077 \\ ? & ? & ? \end{bmatrix}$$

Concept check: write a P matrix



$$P = \begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

Marginal Distribution

- Let's consider a Markov chain $\{X_t\}$ with a transition matrix P .
- Suppose we know the distribution of X_t it is ψ_t .
- How would we get the distribution of X_{t+1} ?

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For any $x_j \in S$

$$P(X_{t+1} = x_j) = \sum_{x_i \in S} P(X_{t+1} = x_j \mid X_t = x_i) \cdot P(X_t = x_i).$$

Therefore,

$$\psi_{t+1}(x_j) = \sum_{x_i \in S} \psi_t(x_i) \cdot P_{ij}$$

We can perform this calculation for all $x_j \in S$ to obtain ψ_{t+1} .

Alternatively:

If ψ_t is a row vector, i.e.,

$$\psi_t = [\psi_t(x_1), \dots, \psi_t(x_n)],$$

then we have

$$\psi_{t+1} = \psi_t P$$

and, for any nonnegative integer s ,

$$\psi_{t+s} = \psi_t P^s.$$

Example:

Recall a transition matrix $\mathbf{P}$ for unemployed/employed:

$$\mathbf{P} = \begin{bmatrix} 0.9 & 0.1 \\ 0.05 & 0.95 \end{bmatrix}$$

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Example continued:

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You see why:

$$\psi_{t+s} = \psi_t P^s.$$

Cross-sectional distributions perspective:

- The ψ_t can be viewed as PMF or cross-sectional frequencies in large samples.
- In unemployment example assume there are many identical workers in the economy:
 - They are independent of each other and all have the same stochastic matrix P .
 - ψ_t is the current cross-sectional distribution over unemployed/employed.
 - i.e., we can interpret $\psi_t(1)$ as the unemployment rate at t .
- The cross-sectional distribution tomorrow is

$$\psi_{t+1} = \psi_t P.$$

The stationary distribution

We are often interested in ψ^* such that:

$$\psi^* = \psi^* P. \quad (1)$$

We call distribution ψ^* *stationary*.

- Note: $\psi^* P$ is a distribution in the next period.
- The Equation (1) requires that ψ^* *doesn't* change over time.
- If the chain starts in $\psi^* P$ it will have such distribution forever!
- We can treat $\psi^* P$ as some long-run “equilibrium”.

How to find the stationary distribution

Under irreducibility and aperiodicity:

$$\psi_0 P^t \rightarrow \psi^*$$

as $t \rightarrow \infty$

- One way: start with (any) ψ_0 , calculate $\psi_t = \psi_0 P^t$, t is (very?) large.
- Other way: solve the Equation (1) with some linear algebra

Approximation

We often assume a continuous stochastic process

For example, consider a linear Gaussian AR(1) model:

$$X_{t+1} = \rho X_t + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_\epsilon^2), \quad |\rho| < 1.$$

- The model has a unique stationary distribution:

$$\psi^* = N(\mu_x, \sigma_x^2), \quad \mu_x := 0, \quad \sigma_x^2 := \frac{\sigma_\epsilon^2}{1 - \rho^2}.$$

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Goal: Find a discrete approximation.

- This means finding a grid X and a transition matrix P such that ψ^* is close to the stationary distribution of the Markov chain with transition matrix P .

Once we do it, we can work with the discrete approximation instead of the continuous model.

There are several ways to do it. For example:

1. Tauchen method (this one we will discuss in more detail)
2. Rouwenhurst method

1. We start by creating the grid X .
2. Pick a number $m > 0$ and an integer N .
3. Set: $x_1 = -m\sigma_x$ and $x_N = m\sigma_x$.

4. Set:

$$x_i = x_1 + \frac{i-1}{N-1} (x_N - x_1) \quad \text{for } i = 2, \dots, N-1.$$

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Note:

$$\text{for } i = 1, \dots, N-1: \quad x_{i+1} - x_i \equiv d$$

d is just a constant.

The space between states is allocated equally:

$$\begin{aligned}X &\in (-\infty, x_1 + 0.5d) \rightarrow x_1 \\X &\in (x_2 - 0.5d, x_2 + 0.5d) \rightarrow x_2 \\&\vdots \\X &\in (x_N - 0.5d, \infty) \rightarrow x_N\end{aligned}$$

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For $j = 1$ and $j = N$:

$$\begin{aligned}P_{i1} &= F(x_1 + 0.5d - \rho x_i) \\P_{iN} &= 1 - F(x_N + 0.5d - \rho x_i)\end{aligned}$$

Tauchen Method (final)

- Usually, we pick m equal to three. Our discrete state space is three times the standard deviation of the stationary distribution in each direction.
- If $\rho \approx 1$ in the AR(1) process, it is difficult to obtain a discrete approximation that matches both the unconditional variance and the first-order auto-correlation, unless the number of grid points is large.
- A wide domain is needed because realizations can be far from the unconditional mean.
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There exist a better alternative: the Rouwenhorst method guarantees to match these 2 moments even with a small number of points.